

PROBLEM STATEMENT #36 | Retail, E-Commerce & Finance

Restaurant Table Booking & Pre-Ordering App

1. Problem Context & Overview

A restaurant hospitality app enabling diners to select specific tables on a live floor plan, place advance food orders, and calculate kitchen preparation timelines for seamless arrival dining.

Target Stakeholders / Actors: *Diner, Restaurant Manager*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.

Sample Acceptance Criteria: Pass: Table reserved and kitchen pre-order ticket queued; Fail: Overbooking of table during overlapping dining window.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.